

WHICH LATTICE MINIMISES A STEEP THREE-BODY POWER-LAW ENERGY? A MAX-MIN TRIANGLE PROBLEM

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ABSTRACT. For a lattice $\Lambda \subset \mathbb{R}^d$ of unit covolume consider the three-body energy $T_\nu(\Lambda) = \sum_{x,y \in \Lambda \setminus \{0\}, x \neq y} (|x||y||x-y|)^{-\nu}$, i.e. the sum over all lattice triangles through the origin of the product of the side lengths to the power $-\nu$. In $d = 2$ this is, up to normalisation, the two-loop modular graph function $C_{s,s,s}(\tau)$ with $s = \nu/2$. We prove that as $\nu \rightarrow \infty$ every family of minimisers of T_ν converges to the rectangular lattice with aspect ratio $\sqrt{(\sqrt{17}-1)/2}$; in particular neither the hexagonal nor the square lattice is optimal for large ν . The proof reduces the limit to a max-min problem for the smallest triangle product of a lattice, which we solve exactly. We complement this with a numerical phase diagram for finite ν (hexagonal $\rightarrow$ square $\rightarrow$ rectangular) and with numerical evidence that in $d = 3$ the body-centred cubic lattice solves the corresponding max-min problem.

1. INTRODUCTION

Lattice energy minimisation for *pair* interactions is a classical subject; for completely monotone pair potentials in the plane the hexagonal lattice is optimal [1, 2, 3, 4, 5], and in higher dimensions universal optimality has been proved in dimensions 8 and 24 [6]. Much less is known for genuine many-body energies. Here we study the simplest one, a product-form three-body power law

$$(1) \quad T_\nu(\Lambda) = \sum_{\substack{x,y \in \Lambda \setminus \{0\} \\ x \neq y}} (|x||y||x-y|)^{-\nu},$$

which converges for $\nu > 2d/3$. This sum (under the name three-body zeta function $\zeta_\Lambda^{(3)}(\nu, \nu, \nu)$) appears as the radial part of the Axilrod–Teller–Muto energy at $\nu = 3$ in [7], where it was evaluated for selected lattices and along the Bain path; efficient algorithms are given in [8, 9]. We stress that (1) is a *model* energy: physical three-body dispersion carries an angular factor, and a pure three-body energy at fixed density does not describe a material. The motivation here is mathematical, see Section 4: in $d = 2$, T_{2s} is a modular graph function.

What is automatic and what is not. T_ν is invariant under isometries, so on the modular surface the hexagonal point $\rho = e^{i\pi/3}$ and the square point i (elliptic fixed points) are always critical points. The content of the results below is *global* minimality and the location of the transitions.

Main result. Let

$$a_* = \left(\frac{1+\sqrt{17}}{8}\right)^{1/4}, \quad P_* = 2a_*^3 = \sqrt{a_*^2 + a_*^{-2}} \approx 1.4317345, \quad y_\infty = a_*^{-2} = \sqrt{\frac{\sqrt{17}-1}{2}} \approx 1.2496211,$$

and let $\Lambda_* = a_*\mathbb{Z} \times a_*^{-1}\mathbb{Z}$ (rectangular, aspect ratio y_∞ , unit covolume).

Theorem 1. *Let $d = 2$. For every $\nu > 4/3$ the energy T_ν attains its minimum on the space of unit-covolume lattices. If Λ_ν is any minimiser, then $\Lambda_\nu \rightarrow \Lambda_*$ up to isometries as $\nu \rightarrow \infty$, and $\lim_{\nu \rightarrow \infty} (\min T_\nu)^{1/\nu} = 1/P_*$. Equivalently (Section 4), minimisers τ_s of the modular graph function $C_{s,s,s}$ on the fundamental domain satisfy $\tau_s \rightarrow i y_\infty$ as $s \rightarrow \infty$.*

Theorem 1 follows from an exact solution of the following max-min problem.

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Definition 2. For a lattice Λ let $\pi(p, q, r) = |p - q| |q - r| |r - p|$ and $P(\Lambda) = \min\{\pi(p, q, r) : p, q, r \in \Lambda \text{ pairwise distinct}\}$ (the minimum exists since Λ is discrete).

Theorem 3. For every lattice $\Lambda \subset \mathbb{R}^2$ of covolume 1, $P(\Lambda) \leq P_*$, with equality if and only if Λ is isometric to Λ_* .

2. PROOF OF THEOREM 3

Lemma 4 (reduced basis). After a rotation and a reflection, every unit-covolume $\Lambda \subset \mathbb{R}^2$ has a basis $u = (a, 0)$, $v = (p, a^{-1})$ with $a = \lambda_1(\Lambda)$, $0 \leq p \leq a/2$ and $|v| \geq a$. Moreover $a^4 \leq 4/3$.

Proof. Lagrange–Gauss reduction; the bound is Hermite’s constant $\gamma_2 = 2/\sqrt{3}$. $\square$

Lemma 5 (upper bound). With the notation of Lemma 4, $P(\Lambda) \leq G(a) := \min\{2a^3, \sqrt{a^2 + a^{-2}}\}$, and $\pi(0, u, v) \leq \sqrt{a^2 + a^{-2}}$ with equality iff $p = 0$.

Proof. The collinear triple $(0, u, 2u)$ gives $\pi = a \cdot a \cdot 2a = 2a^3$. For the triple $(0, u, v)$, $\pi(0, u, v)^2 = a^2(p^2 + k)((a - p)^2 + k)$ with $k = a^{-2}$. Put $m = p(a - p) \in [0, a^2/4]$. Since $p^2 + (a - p)^2 = a^2 - 2m$,

$$(p^2 + k)((a - p)^2 + k) = m^2 + k(a^2 - 2m) + k^2 = (ka^2 + k^2) + m(m - 2k).$$

Because $a^4 \leq 4/3 < 8$ we have $m \leq a^2/4 < 2a^{-2} = 2k$, hence $m(m - 2k) \leq 0$ with equality iff $m = 0$, i.e. $p = 0$ (note $p \leq a/2 < a$). Therefore $\pi(0, u, v)^2 \leq a^2(ka^2 + k^2) = a^2 + a^{-2}$. $\square$

Lemma 6. For $0 < a \leq (4/3)^{1/4}$ we have $G(a) \leq P_*$, with equality iff $a = a_*$.

Proof. a_* is the positive root of $4a^8 = a^4 + 1$, equivalently $2a_*^3 = \sqrt{a_*^2 + a_*^{-2}}$. For $a \leq a_*$, $G(a) \leq 2a^3 \leq 2a_*^3$ with equality iff $a = a_*$. The function $g(a) = a^2 + a^{-2}$ is strictly decreasing on $(0, 1]$ and increasing on $[1, \infty)$. For $a_* < a \leq 1$, $G(a) \leq \sqrt{g(a)} < \sqrt{g(a_*)} = P_*$. For $1 \leq a \leq (4/3)^{1/4}$, $g(a) \leq g((4/3)^{1/4}) = \frac{2}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{7}{2\sqrt{3}}$, and $(\frac{7}{2\sqrt{3}})^2 = \frac{49}{12} < 16(\frac{1+\sqrt{17}}{8})^3 = P_*^4$ (numerically $4.083 < 4.202$), so again $G(a) < P_*$. $\square$

Lemma 7 (the extremal lattice). $P(\Lambda_*) = P_*$.

Proof. Write $u = (a_*, 0)$, $v = (0, a_*^{-1})$; note $a_* < 1$. The triples $(0, u, 2u)$ and $(0, u, v)$ give $\pi = P_*$, so $P(\Lambda_*) \leq P_*$. *Collinear triples.* Three distinct lattice points on a line are $q + jsw$, $q + ksw$, $q + lsw$ with w a unit vector, $s \geq \lambda_1 = a_*$ and integers $j < k < l$; then $\pi = s^3(k - j)(l - k)(l - j) \geq 2s^3 \geq 2a_*^3 = P_*$. *Non-collinear triples.* Let the side lengths be $d_1 \leq d_2 \leq d_3$ and γ the angle between the sides d_1, d_2 . The triangle has area $\geq 1/2$, so $d_1 d_2 \geq d_1 d_2 \sin \gamma = 2 \text{ area} \geq 1$ and $\pi \geq d_3$. We claim $d_3 \geq D := \sqrt{a_*^2 + a_*^{-2}} = P_*$. The three side vectors are nonzero, pairwise linearly independent vectors of Λ_* . A vector (ma_*, na_*^{-1}) has norm $< D$ only if $(m, n) \in \{(\pm 1, 0), (0, \pm 1)\}$: if $n \neq 0$ then $m^2 a_*^2 < a_*^2 + (1 - n^2) a_*^{-2} \leq a_*^2$ forces $m = 0$ and $n^2 < 1 + a_*^4 < 2$; if $n = 0$ then $m^2 < 1 + a_*^{-4} = 1 + y_\infty^2 < 4$. These vectors span only two directions, so at most two of the three pairwise independent side vectors can be shorter than D . Hence $d_3 \geq D$ and $\pi \geq P_*$. $\square$

Proof of Theorem 3. By Lemmas 4–6, $P(\Lambda) \leq G(a) \leq P_*$. If $P(\Lambda) = P_*$, then $G(a) = P_*$, so $a = a_*$; moreover $\pi(0, u, v) \geq P(\Lambda) = P_* = \sqrt{a_*^2 + a_*^{-2}}$, which by Lemma 5 forces $p = 0$, i.e. $\Lambda = \Lambda_*$. Conversely $P(\Lambda_*) = P_*$ by Lemma 7. $\square$

3. PROOF OF THEOREM 1

Let $\mathcal{L}_2$ be the space of unit-covolume lattices modulo isometries, with the topology induced by bases.

Lemma 8. (a) $T_\nu(\Lambda) \geq P(\Lambda)^{-\nu}$ for all Λ and $\nu > 4/3$. (b) P is continuous on $\mathcal{L}_2$ and $P(\Lambda) \leq 2\lambda_1(\Lambda)^3$. (c) T_ν is lower semicontinuous on $\mathcal{L}_2$. (d) For $\nu \geq \nu_0 > 4/3$: $T_\nu(\Lambda_*) \leq C P_*^{-\nu}$ with $C = T_{\nu_0}(\Lambda_*) P_*^{\nu_0}$.

Proof. (a) Translating a minimal triple so that one vertex is 0 produces a term $P(\Lambda)^{-\nu}$ of (1). (b) The bound is the triple $(0, u, 2u)$. Continuity: near a given lattice only triples of points in a fixed ball are relevant, and π is continuous in the basis. (c) T_ν is a sum of nonnegative continuous functions of the basis. (d) Every term satisfies $\pi \geq P_*$, so $(P_*/\pi)^\nu \leq (P_*/\pi)^{\nu_0}$. $\square$

Proof of Theorem 1. Existence. By Lemma 8(a),(b), $T_\nu(\Lambda) \geq (2\lambda_1^3)^{-\nu}$, so sublevel sets of T_ν lie in $\{\lambda_1 \geq \varepsilon\}$, which is compact by Mahler's criterion; with (c) a minimiser exists. *Convergence.* For a minimiser Λ_ν and $\nu \geq \nu_0$, Lemma 8 gives $P(\Lambda_\nu)^{-\nu} \leq T_\nu(\Lambda_\nu) \leq T_\nu(\Lambda_*) \leq CP_*^{-\nu}$, hence $P_* \geq P(\Lambda_\nu) \geq C^{-1/\nu}P_* \rightarrow P_*$. In particular $\lambda_1(\Lambda_\nu)$ is bounded below, so (Λ_ν) lies in a compact set. Any limit point Λ_∞ satisfies $P(\Lambda_\infty) = P_*$ by continuity, hence $\Lambda_\infty \cong \Lambda_*$ by Theorem 3. The same chain of inequalities gives $(\min T_\nu)^{1/\nu} \rightarrow 1/P_*$. $\square$

4. RELATION TO MODULAR GRAPH FUNCTIONS

For τ in the upper half plane the two-loop modular graph function is $C_{a,b,c}(\tau) = \sum_{p_1+p_2+p_3=0, p_i \neq 0} \prod_i \left(\frac{\text{Im } \tau}{\pi |p_i|^2}\right)^a$ $p_i \in \mathbb{Z} + \tau\mathbb{Z}$ [10]. With $\Lambda_\tau = (\mathbb{Z} + \tau\mathbb{Z})/\sqrt{\text{Im } \tau}$ (unit covolume) one has

$$C_{s,s,s}(\tau) = \pi^{-3s} T_{2s}(\Lambda_\tau),$$

so Theorem 1 is a statement about the location of the minimum of $C_{s,s,s}$ on the fundamental domain. At $s = 1$, Zagier's identity $C_{1,1,1} = E_3 + \zeta(3)$ and the minimality of the Epstein zeta function at ρ [1, 2, 3, 4] show that the minimum is at ρ (hexagonal). Theorem 1 shows that for large s it is near iy_∞ . We are not aware of earlier results on the minima of $C_{s,s,s}$ for $s > 1$; see the caveat in Section 7.

5. NUMERICAL PHASE DIAGRAM FOR FINITE ν

The following are numerical observations, not theorems. T_ν was evaluated by cube-truncated direct summation with an FFT convolution and Richardson extrapolation in the cut-off (independent of [9]); agreement with the library of [9] is better than $2 \cdot 10^{-10}$ relative on 55 test cases, and the values at $\nu = 3$ reproduce those of [7] after conversion to unit covolume.

- $2 < \nu < \nu^*$: the hexagonal lattice is the global minimiser. At $\nu = 2$ this is exact by Section 4.
- $\nu^* = 3.9183649026\dots$: first-order transition to the square lattice; both are local minima for $3.806 < \nu < 4.278$.
- $\nu_2 \approx 8.606$: the square lattice becomes unstable towards rectangular deformations; the optimal aspect ratio then grows continuously (1.056 at $\nu = 9$, 1.100 at 10, 1.141 at 12, 1.170 at 15, 1.194 at 20, 1.215 at 30), consistent with the limit $y_\infty = 1.2496$ of Theorem 1.

Conjecture 9. For $d = 2$ the minimiser of T_ν is hexagonal for $4/3 < \nu < \nu^*$, square for $\nu^* < \nu \leq \nu_2$ and rectangular with aspect ratio increasing from 1 to y_∞ for $\nu > \nu_2$.

6. THREE DIMENSIONS

In $d = 3$ the collinear triple gives again $P(\Lambda) \leq 2\lambda_1^3$. For the body-centred cubic lattice (unit covolume) one finds $P(\text{BCC}) = 3/2$, attained by the isosceles triangles formed by two nearest neighbours and a second neighbour, while $P(\text{FCC}) = P(\text{SC}) = \sqrt{2}$. A numerical maximisation of P over the five-dimensional space of 3D lattices (62 Nelder–Mead runs) never exceeded $3/2$ and returned BCC in the best runs.

Conjecture 10. For unit-covolume lattices in $\mathbb{R}^3$, $P(\Lambda) \leq 3/2$ with equality only for BCC. Consequently (by the argument of Theorem 1) minimisers of T_ν converge to BCC as $\nu \rightarrow \infty$.

Numerically, BCC is the best lattice found for all tested $\nu \in [3.2, 12]$, and FCC ceases to be a local minimum near $\nu \approx 3.75$ (preliminary; error bars pending).

7. LIMITATIONS

(i) T_ν is a model energy; physically relevant three-body dispersion has an angular factor and is always accompanied by pair terms. (ii) Non-Bravais structures such as hcp are not covered by the lattice setting. (iii) Apart from Theorems 1 and 3, all statements are numerical. (iv) Our literature search for prior results on minima of $C_{s,s,s}$ and on minimisers of T_ν was not exhaustive.

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